

HYEONAH KIM

Assistant Professor
Ulsan National Institute of Science and Technology (UNIST)

Department of Industrial Engineering
Graduate School of Artificial Intelligence

✉ hyeonah.kim@unist.ac.kr
🌐 hyeonahkimm.github.io
🔄 github.com/hyeonahkimm
🌐 linkedin.com/in/hyeonahkimm

RESEARCH INTERESTS

- Generative models, reinforcement learning, and search for **structured and discrete decision spaces**
- **AI for scientific discovery and engineering**, including molecular and biological design, combinatorial optimization, and black-box optimization

ACADEMIC & PROFESSIONAL EXPERIENCE

9/2026 – Present	Assistant Professor Department of Industrial Engineering Joint appointment: Graduate School of Artificial Intelligence Structured Discovery Lab	UNIST
4/2025 – 8/2026	Postdoctoral Researcher Supervisors: Alex Hernández-García Generative models, active learning, and off-policy reinforcement learning for scientific discovery; re-researcher in the PandemicStop-AI consortium	Mila / Université de Montréal
1/2015 – 6/2017	Software Engineer LGE ERP Manufacturing; database engineering (Oracle PL/SQL)	LG CNS

EDUCATION

3/2021 – 2/2025	Ph.D. in Industrial and Systems Engineering Thesis: <i>Algorithmic Priors for Neural Combinatorial Optimization</i> Supervisor: Jinkyoo Park; Co-supervisor: Changhyun Kwon	KAIST
3/2019 – 2/2021	M.S. in Industrial Engineering Optimization and Operational Research Lab.; Supervisor: Sung-Pil Hong	Seoul National University
3/2011 – 2/2015	B.S. in Industrial Engineering, Summa Cum Laude Information Design Lab.	Hanyang University

PUBLICATIONS

ICML, 2026	[11] Synthesizable Molecular Generation via Soft-constrained GFlowNets with Rich Chemical Priors [paper][code] Hyeonah Kim , Minsu Kim, Celine Roget, Dionesa Biton, Louis Vaillancourt, Yves V. Brun, Yoshua Bengio, Alex Hernández-García
Preprint, 2025	[10] Test-Time Search in Neural Graph Coarsening Procedures for the Capacitated Vehicle Routing Problem [paper] Yoonju Sim, Hyeonah Kim , Changhyun Kwon
KDD, 2025	[9] RL4CO: An Extensive Reinforcement Learning for Combinatorial Optimization Benchmark [paper][code] Federico Berto*, Chuanbo Hua*, Junyoung Park*, Laurin Luttman*, Yining Ma, Fanchen Bu, Jiarui Wang, Haoran Ye, Minsu Kim, Sanghyeok Choi, Nayeli Gast Zepeda, André Hottung, Jianan Zhou, Jieyi Bi, Yu Hu, Fei Liu, Hyeonah Kim , Jiwoo Son, Haeyeon Kim, Davide Angioni, Wouter Kool, Zhiguang Cao, Qingfu Zhang, Joungho Kim, Jie Zhang, Kijung Shin, Cathy Wu, Sungsoo Ahn, Guojie Song, Changhyun Kwon, Kevin Tierney, Lin Xie, Jinkyoo Park <i>Datasets and Benchmarks Track, Oral</i>
ICML, 2025	[8] Neural Genetic Search in Discrete Spaces [paper][code] Hyeonah Kim* , Sanghyeok Choi*, Jiwoo Son, Jinkyoo Park, Changhyun Kwon
ICML, 2025	[7] Improved Off-policy Reinforcement Learning in Biological Sequence Design [paper][code] Hyeonah Kim , Minsu Kim, Taeyoung Yun, Sanghyeok Choi, Emmanuel Bengio, Alex Hernández-García, Jinkyoo Park
AISTATS, 2025	[6] Ant Colony Sampling with GFlowNets for Combinatorial Optimization [paper][code] Minsu Kim*, Sanghyeok Choi*, Hyeonah Kim , Jiwoo Son, Jinkyoo Park, Yoshua Bengio

NeurIPS, 2024	[5] Genetic-guided GFlowNets for Sample Efficient Molecular Optimization [paper][code] Hyeonah Kim , Minsu Kim, Sanghyeok Choi, Jinkyoo Park
ICML, 2024	[4] Symmetric Replay Training: Enhancing Sample Efficiency in Deep Reinforcement Learning for Combinatorial Optimization [paper][code] Hyeonah Kim , Minsu Kim, Sungsoo Ahn, Jinkyoo Park
AAAI, 2024	[3] Equity-Transformer: Solving NP-hard Min-max Routing Problems as Sequential Generation with Equity Context [paper] Jiwoo Son*, Minsu Kim*, Sanghyeok Choi, Hyeonah Kim , Jinkyoo Park
IJOC, 2024	[2] A Neural Separation Algorithm for the Rounded Capacity Inequalities [paper] Hyeonah Kim , Jinkyoo Park, Changhyun Kwon <i>INFORMS Journal on Computing</i> , 36(4), 987-1005
ICML, 2023	[1] Meta-SAGE: Scale Meta-Learning Scheduled Adaptation with Guided Exploration for Mitigating Scale Shift on Combinatorial Optimization [paper] Jiwoo Son*, Minsu Kim*, Hyeonah Kim , Jinkyoo Park

* Equal contribution.

HONORS AND AWARDS

9/2025 – 8/2026	Sejong Science Fellowship 2025 Young Scientist Grant Program	National Research Foundation of Korea
12/2024	Google Conference Scholarship NeurIPS 2024; first-author paper: <i>Genetic-guided GFlowNets for Sample Efficient Molecular Optimization</i>	Google Asia
10/2024	KAIST Graduate Student Outstanding Paper Award 2024 First-author paper: <i>A Neural Separation Algorithm for the Rounded Capacity Inequalities</i>	KAIST